How the Web Works

From the address bar to the first page

`https` `://` `example.com` `:443` `/docs` `?page=2` `#intro`

scheme · host · port · path · query · fragment

Sent: **path, query, Host**. Not sent: **#fragment**.

After you press Enter

1. **DNS** — name to IP
2. **TCP** — open connection
3. **TLS** — verify, encrypt
4. **HTTP** — request, response
5. **Render** — parse, fetch more

1 Browser asks

```
GET /docs?page=2 HTTP/1.1
Host: example.com
Accept: text/html
// then waits for the reply
```

2 Server answers

```
HTTP/1.1 200 OK
Content-Type: text/html
<link href="/app.css">
<script src="/app.js">
// then GET /app.css, /app.js
```

TRAP What happens here?

```
// Press Enter. The server gets?
"https://shop.dev:8443/cart?id=7#top"
```

```
GET /cart?id=7
Host: shop.dev:8443
```

#top never leaves the browser.

KEY TAKEAWAY

- **DNS** — Name to IP
- **TCP + TLS** — Connect, secure
- **HTTP** — Then ask

? Follow-up questions to expect

- **What does DNS return?** → An IP address, cached until its TTL runs out.
- **Why TLS on top of TCP?** → TCP only moves bytes. TLS adds encryption and identity.
- **Is the `#fragment` sent?** → No. Only the browser reads it.
- **What changes in HTTP/3?** → QUIC over UDP, with TLS 1.3 built in.